# Vector control

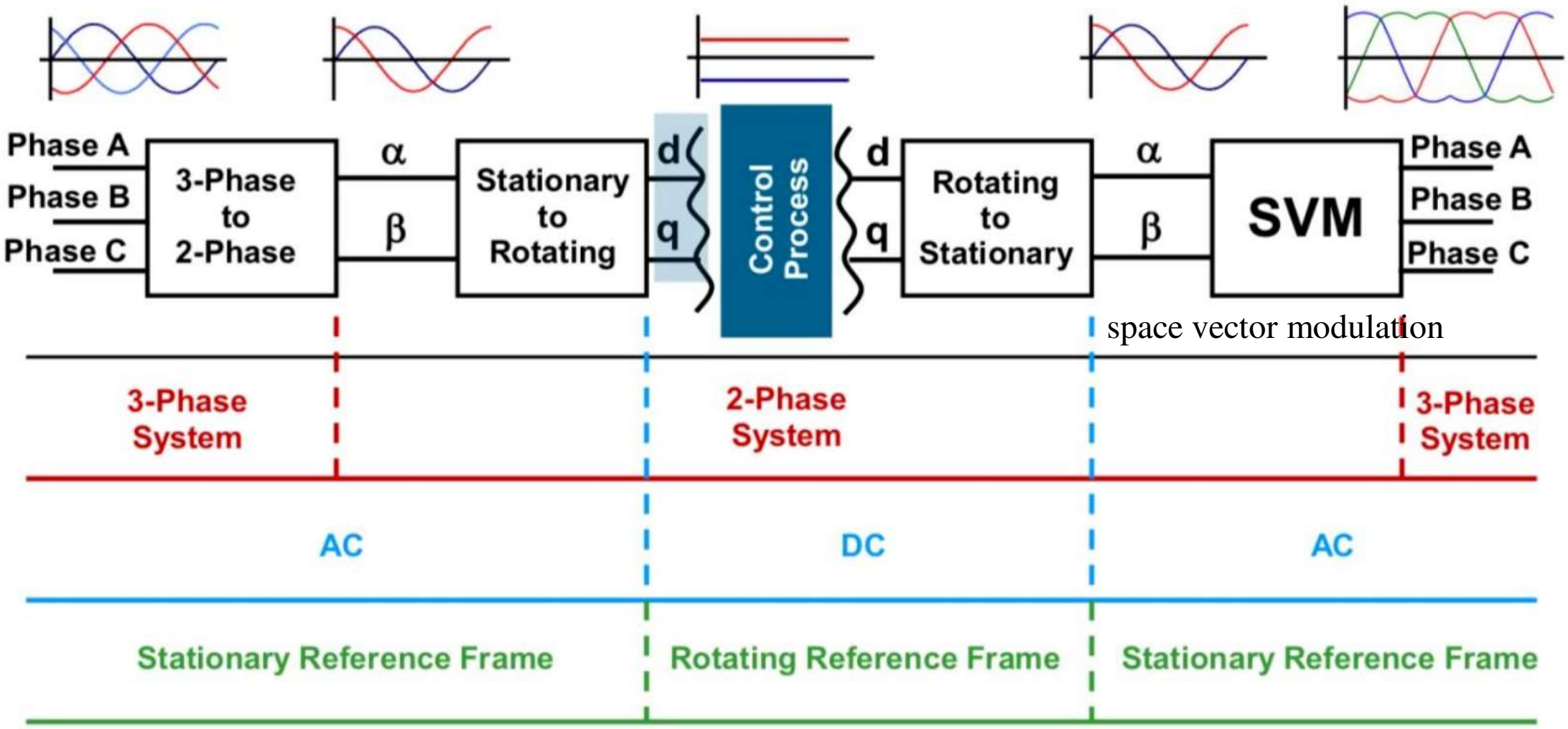
Vector control — also referred to as field oriented control (FOC) — controls the speed or torque of an AC motor by controlling the stator current space vectors, in manner similar to (but more complicated than) DC control methods. Field oriented control uses complex mathematics to transform a three-phase system that depends on time and speed to a two coordinate (d and q) time-invariant system.

The stator current in an AC motor is made up of two components: the magnetizing component (d) of the current and the torque-producing component (q). With FOC, these two current components are controlled independently. This allows the torque-producing component, q, to be kept orthogonal to the rotor flux for maximum torque production, and therefore, optimum speed control.

Like scalar methods, vector control can be open-loop or closed-loop. Open-loop vector control (also referred to as senseless vector control) uses a mathematic model of the motor operating parameters, rather than using a physical feedback device. The controller monitors voltage and current from the motor and compares them to the mathematical model. It then corrects any errors by adjusting the current supplied to the motor, which adjusts the motor's torque production accordingly ($T = K\varphi I_2' \cos \phi_2$). With senseless vector control, it's important to have a very accurate mathematical model of the motor, and the controller must be tuned for proper operation. Closed-loop vector control uses an encoder to provide shaft position feedback, and this information is sent to the controller, which adjusts the supplied voltage to increase or decrease torque. This is the only method that allows direct torque control in all four quadrants of motor operation for dynamic braking or regeneration.

Vector control methods are more complex than scalar VFD control methods, but they offer significant benefits over scalar methods in some applications. For example, open-loop vector control enables the motor to produce high torque at low speeds, and closed-loop vector control allows a motor to produce up to 200 percent of its rated torque at zero speed, useful for holding loads at standstill (مفيد للامساك بالاحمال في حالة التوقف التام). Closed-loop vector control also provides very accurate optimum speed control.

## **block-diagram of field –oriented control** SVM- space vector modulation



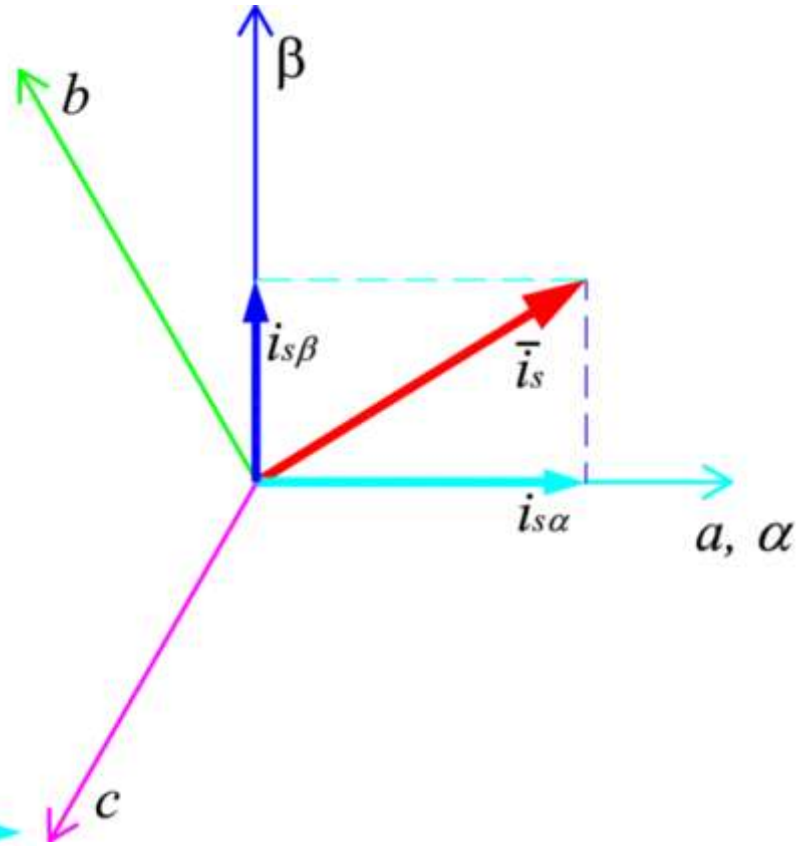
# Clarke transformation

This transformation block is responsible for translating three axes to
two axes system reference to the stator. Two of the three phase currents are measured because the sum of the three phase currents equal to zero. Basically the transformation shift from a three axis, two- dimensional coordinate system attached to the stator of the motor to a two axis system referred to the stator. The measured current represents the vector component of the current in a three axis coordinate system which are spatially separated by 120°. Clarke transformation transforms the rotating current vector in a two axis orthogonal coordinate system, so that the current vector is represented with two vector components which vary with time. The space vector can be transformed to another reference frame with
only two orthogonal axis called α-β, where the axis 'α' and axis '*a*' coincide each other. Fig. 3.15 shows the stator current space vector and its component in stationary reference frame and the Clarke transformation module.
The projection that modifies the three phase system into two dimensional orthogonal system is expressed as:

# Clarke transformation

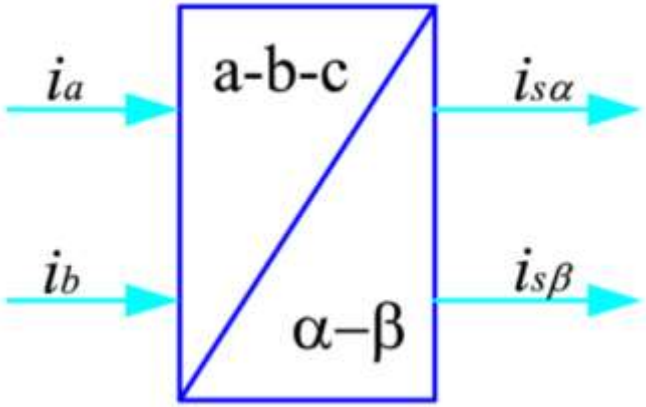
$$i_{s\alpha} = \mathrm{Re}\left( \frac{2}{3}\left( i_a(t) + i_b(t)e^{j2\pi/3} + i_c(t)e^{-j2\pi/3} \right) \right)$$

$$i_{s\beta} = \mathrm{Im}\left( \frac{2}{3}\left( i_a(t) + i_b(t)e^{j2\pi/3} + i_c(t)e^{-j2\pi/3} \right) \right)$$




In metric's form:

$$\begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix}\begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$



# Park transformation

In this transformation, the current phasor is transformed into two components of a rotating reference frame rotating at the same speed as the angular frequency of the phase currents, these components do not depend on time and speed.

The component of stator current which is responsible for the rotor flux can be fix to the d axis ($i_{sd}$) . These components depend o n the α-β current vector components and the rotor flux position. The separate flux and the torque components of stator current vector in two coordinate time invariant system can be expressed by $\vec{i}_s = i_{sd} + j i_{sq}$

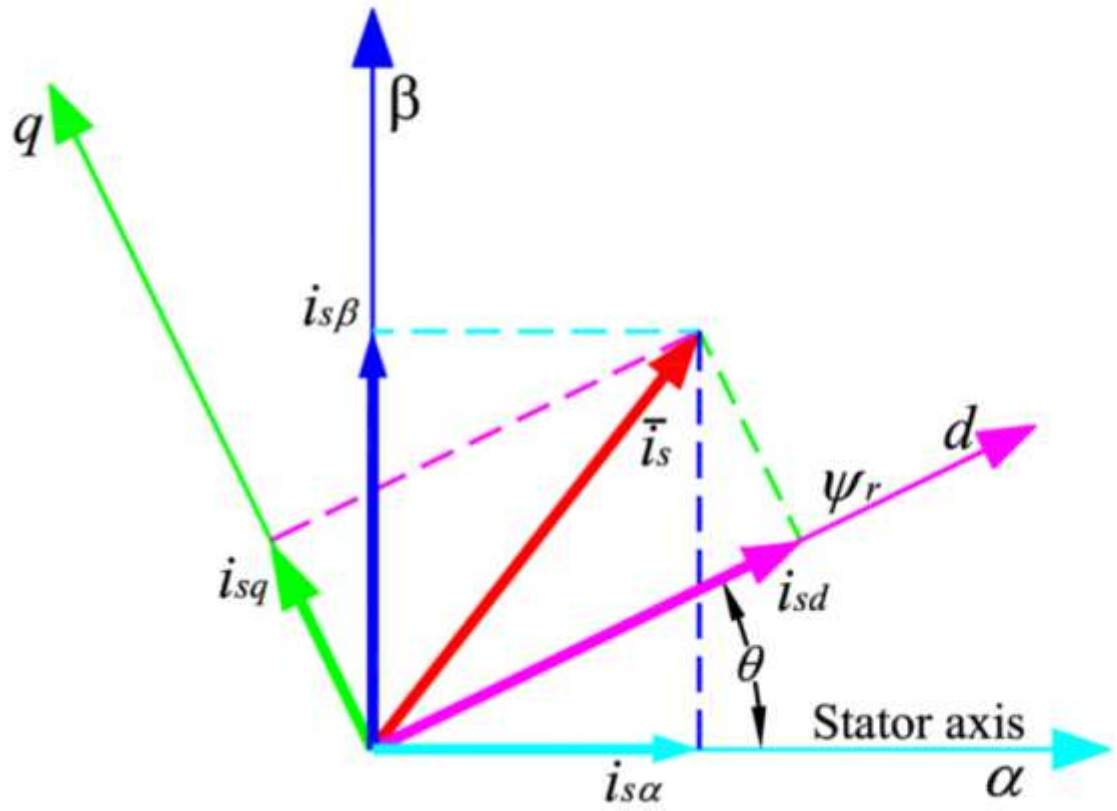
where,

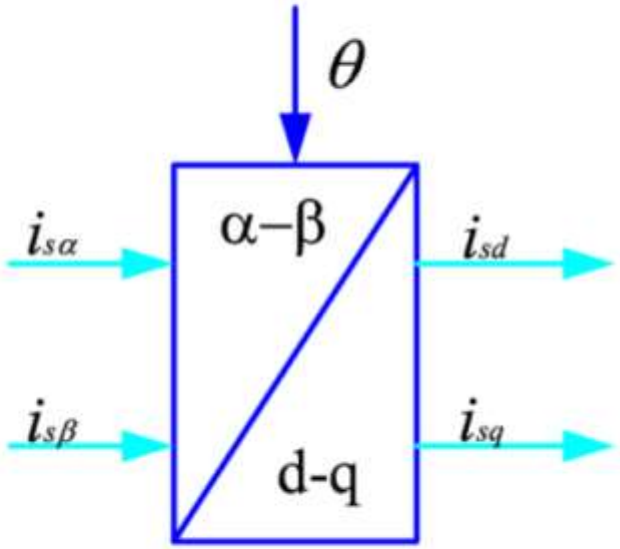
$$\begin{bmatrix} i_{sd} \\ i_{sq} \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \end{bmatrix}$$

Direct torque control is possible and becomes easy with the flux component $i_{sd}$ aligned with the d axis representing the direction of the rotor flux and torque component $i_{sq}$ aligned with the q axis perpendicular to the rotor flux.

Fig. 3.16 shows the stator current space vector and its component in rotating reference frame and the Park transformation module

# Park transformation




Park transformation

Stator current space vector in rotating reference frame

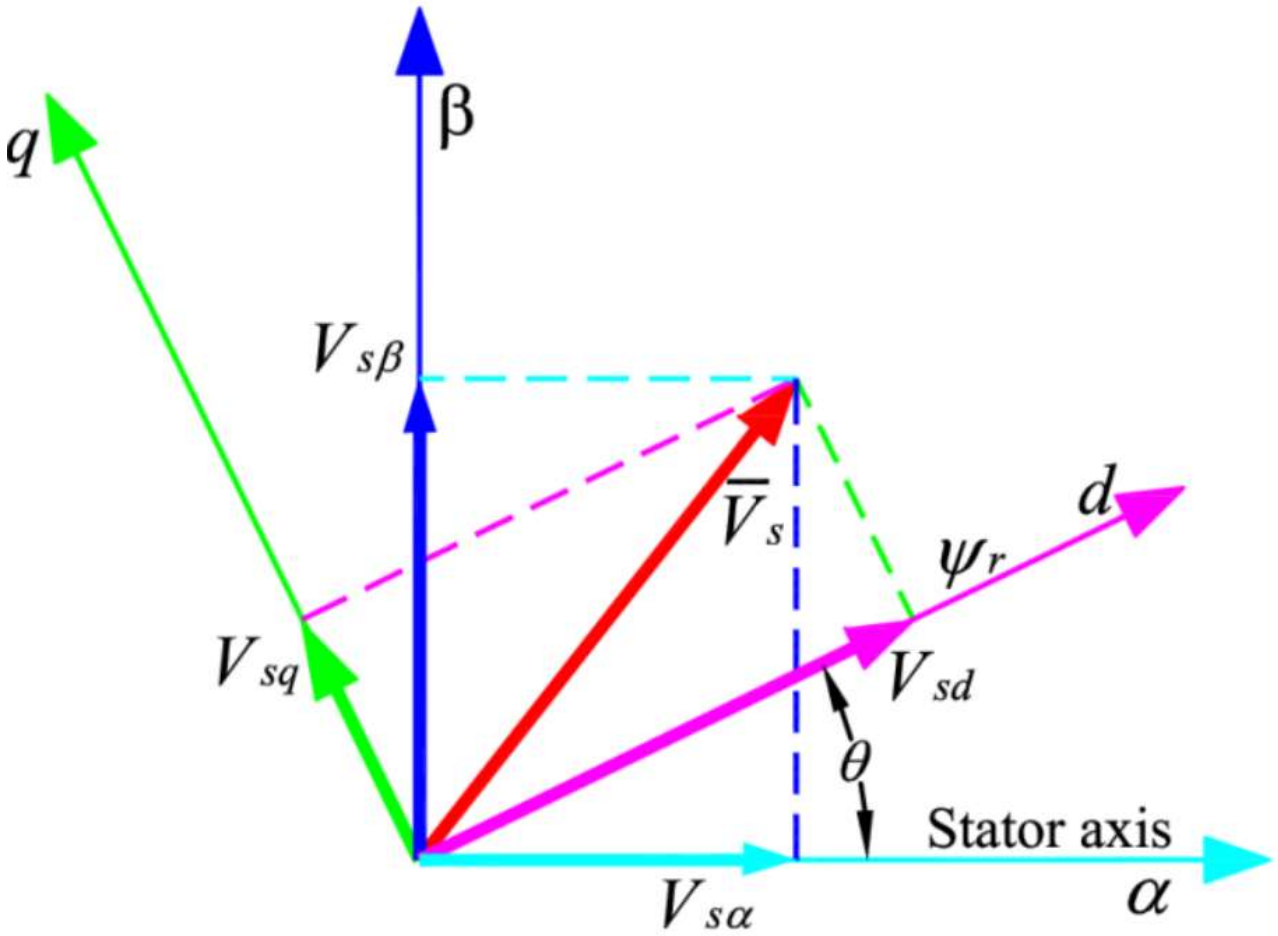
# Inverse Park transformation

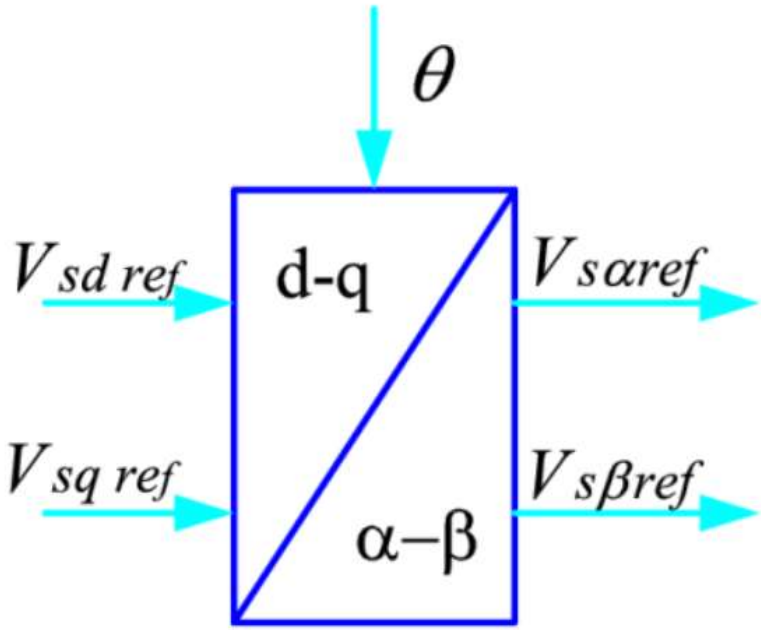
In this transformation, the stator voltages represented in the d-q rotating reference frame are transformed to a two phase orthogonal α-β system, from which we can obtain the reference vector components to be applied to the motor phases through space vector modulation technique. The projection that modifies the d-q rotating reference frame to two phase orthogonal system is expressed by

$$\begin{bmatrix} V_{s\alpha} \\ V_{s\beta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} V_{sd} \\ V_{sq} \end{bmatrix}$$

The following figure shows the Inverse Park transformation module and the stator voltage space vector and its component in stationary orthogonal system.

# Inverse Park transformation




Stator voltage space vector from *d-q* to *α-β*

# Determination of the Rotor Flux Angle (θ)

In induction motor, mechanical speed is defined as the difference between rotor flux speed and slip angular speed and rotor flux linkage can be with the physical quantity as:

$$\omega_r = \omega_e - \omega_{slip}$$

To determine the rotor flux angle(θ), first we need to calculate the rotor flux speed ($\omega_e$) using the following equation:

$$\omega_e = \omega_r + \omega_{slip} = \omega_r + \frac{1}{\tau_r}\frac{i_{sq}}{i_{mr}}$$

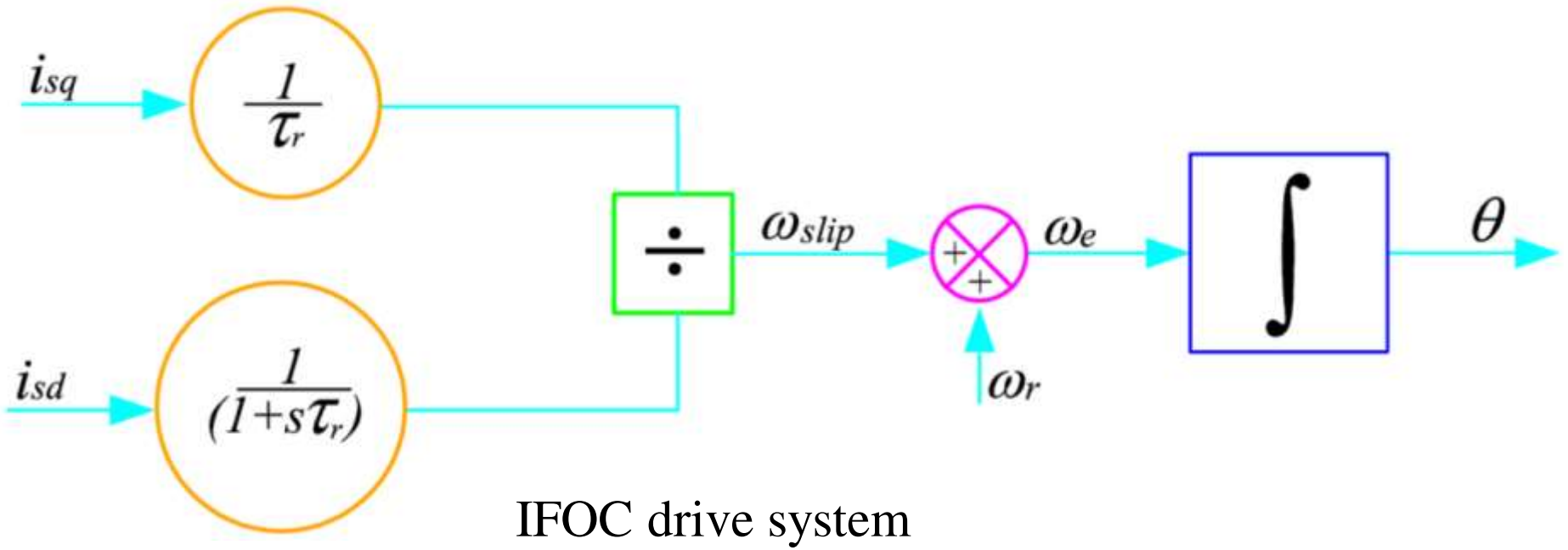
then $\theta = \int \omega_e dt$

where, the motor time constant $\tau_r = \frac{L_r}{R_r}$

the magnetizing current $i_{mr} = \frac{i_{sd}}{(1 + s\tau_r)}$

# Determination of the Rotor Flux Angle (θ)

The flux and torque can be controlled independently by providing the slip frequency
The block diagram of indirect field oriented control (IFOC) drive system is shown in the figure below




IFOC drive system

# Properties of the FOC methods

Properties of the FOC methods are,

i. It is based on the analogy to the control of separately excited DC motor,
ii. Coordinate transformations ( Clarke and Park) are required,
iii. PWM algorithm is needed,
iv. Current controllers are necessary,
v. Sensitive to rotor time constant ($\tau_r$),
vi. Rotor flux estimator is essential in DFOC,
vii. Mechanical speed ($\omega_r$) is required in IFOC.

# Implementation of FOC in Induction Motor Drive System

The main aspect of field oriented control method is the coordinate transformation. Basic Field Oriented induction motor drive system is shown in Fig. 3.18. The current vector is measured in stationary reference frame α-β, where the components of currents $i_{s\alpha}$ and $i_{s\beta}$ must be transformed to the rotating co-ordinate system d-q known as Park transformation.

Similarly the reference stator voltage vector components $V_{s\alpha}$ and $V_{s\beta}$ must be transformed from the d-q system to α-β known as inverse transformation. These transformations require a rotor flux angle ‘θ’. The transition from the stationary reference frame to the rotor rotating reference frame requires the determination of position of rotor. The position estimation can be done through sensorless control. In sensorless control, an estimator block is needed. Two of the three phase currents are measured because the sum of the three phase currents is equal to zero.

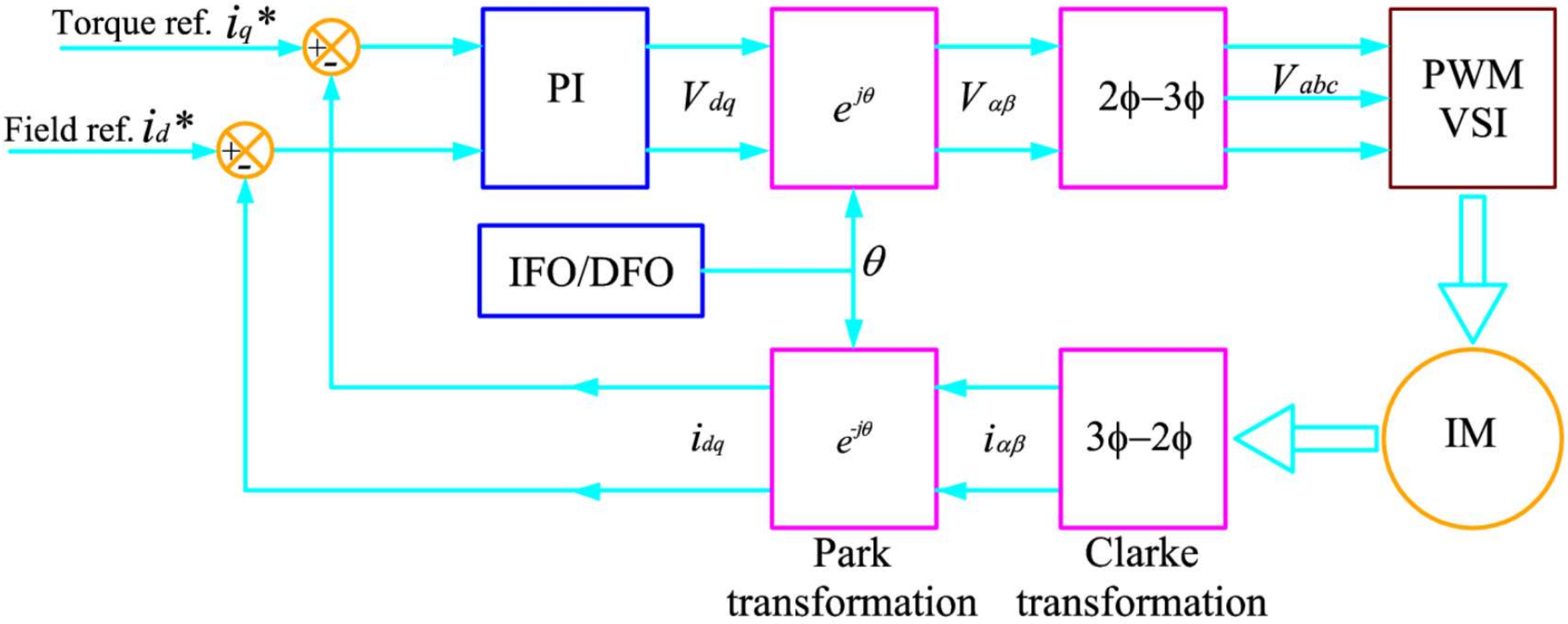



Fig. 3.18 Field oriented Induction motor drive system

This current is fed to the Clarke transformation module, the output obtained from this block is designated as $i_{s\alpha}$ and $i_{s\beta}$. These two components of current act as the input to the Park transformation block, which gives current in the rotating reference frame. Calculation of the two components in the rotating reference frame $i_{sd}$ and $i_{sq}$ is possible by finding the exact rotor flux angular position ($\theta$). These components of currents are compared with the flux reference current $i_{sd},r_{ef}$ and torque reference current $i_{sq},r_{ef}$. The portability from asynchronous to synchronous drive can be obtained by simply changing the flux reference and determining the rotor flux position. The torque command $i_{sq},r_{ef}$ is obtained from the speed regulator output. The output of the current regulators are $V_{sd},r_{ef}$ and $V_{sq},r_{ef}$, they are acting as the input to the inverse Park transformation, where the conversion from d-q to $\alpha$-$\beta$ takes place. The output of this projection gives the component of the stator vector voltage in the $\alpha$-$\beta$ stationary reference frame as $V_{s\alpha},r_{ef}$ and $V_{s\beta},r_{ef}$. The rotor flux position is necessary for Park and Inverse Park transformations. Here space vector modulation techniques are used, which is a sophisticated PWM method that provides advantages such as higher DC bus voltage utilization and lower total harmonic distortion.

# Simulation Model of FOC Induction Motor Drive System

In this field oriented control simulation model, a 1.5 kW induction motor is used, where three phase voltages are converted into two phase reference frame voltages using Clarke transformation module. Park transformation is used to obtain the voltages. From these voltages, associated flux and current are calculated and then applied to electromechanical torque equations to obtain torque speed responses. Fig. 3.19 shows the system configuration of IFOC induction machine with sensor and the simulation is implemented using MATLAB/Simulink.

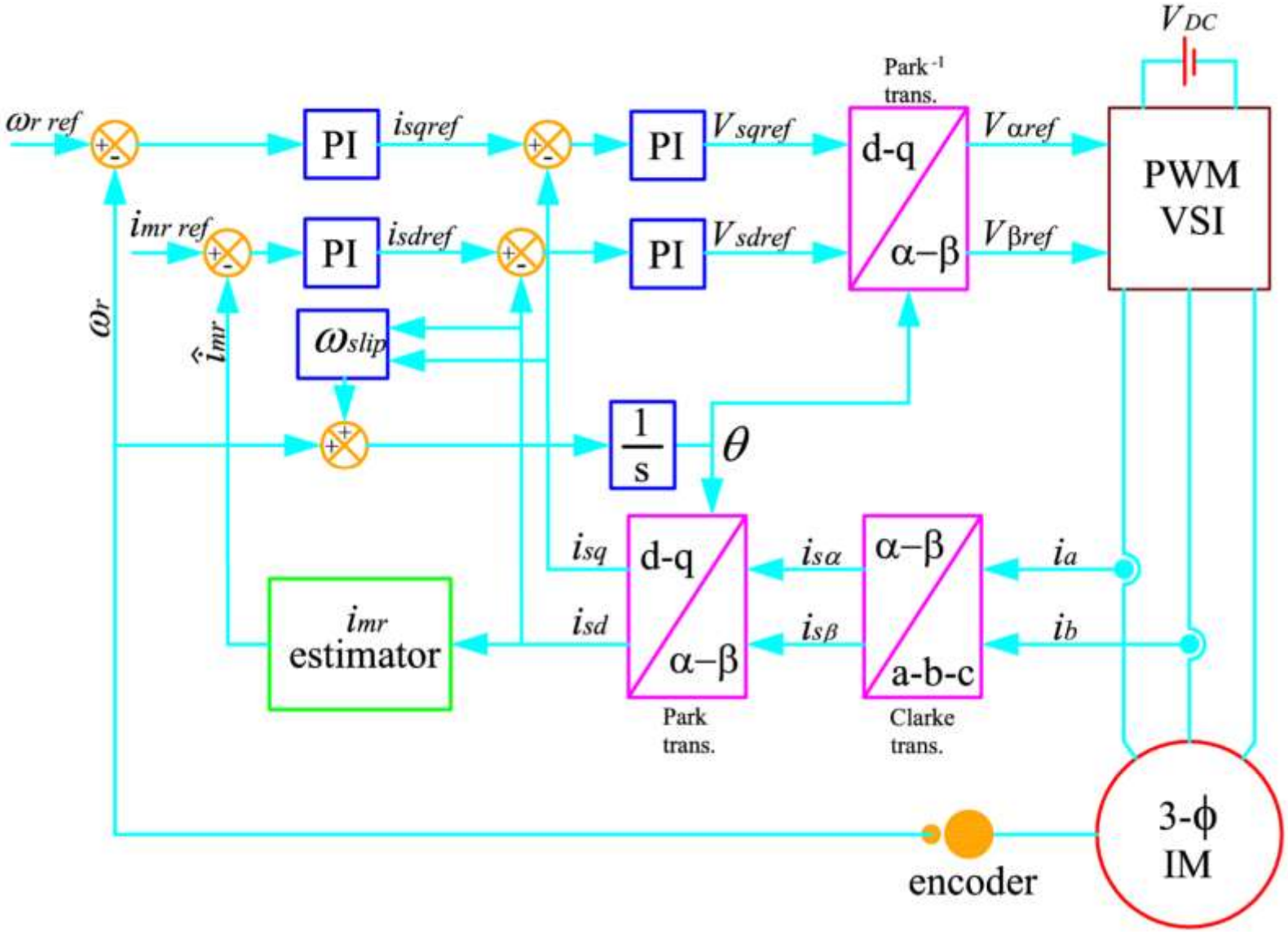



Fig. 3.19 System configuration of IFOC induction machine with sensor

# Comparison between Scalar Control and Vector control of AC motor speed (especially Induction Motors):

## 1. Basic Principle

| Feature | Scalar Control (V/f Control) | Vector Control (Field-Oriented Control - FOC) |
|---|---|---|
| **Control Type** | Controls magnitude of voltage & frequency (V/f ratio) | Controls both magnitude & angle of voltage/current vectors |
| **Decoupling** | No decoupling between flux & torque | Torque and flux are controlled independently |
| **Model-Based** | Open-loop, less mathematical modeling | Requires motor mathematical model |

## 2. Performance

| Feature | **Scalar Control** | **Vector Control** |
|---|---|---|
| **Dynamic Response** | Slower and less accurate | Fast, precise torque and speed control |
| **Torque Control** | Indirect and not very accurate | Direct and precise |
| **Speed Regulation** | Poor at low speed | Excellent at all speeds |
| **Load Disturbance** | Poor rejection | Excellent disturbance rejection |

## 3. Complexity & Cost

| Feature | **Scalar Control** | **Vector Control** |
|---|---|---|
| **Implementation** | Simple and low-cost | Complex with higher processing requirements |
| **Sensors** | Often sensorless | May require position/speed sensors (or estimators) |
| **Hardware** | Basic controllers | Requires DSPs or microcontrollers |

## 4. Applications

**Scalar Control**

HVAC, pumps, fans, conveyors (low-performance)

**Vector Control**

Robotics, CNC, electric vehicles (high-performance)